

Rewards Agent

Personalized rewards optimization via typed-graph multi-agent coordination

PRODUCT PLANNING BRIEF

2-week sprint · live demo June 29
Owner: Raq (team lead) · Team of 4

THE PROBLEM WE ARE SOLVING

Americans earned \$47.5B in card rewards in 2024 and left ~\$27.9B unredeemed. The cause is structural: optimizing a points portfolio is a multi-step planning problem — balances, transfer ratios, award availability, fees, status, and time horizon all interact — not a lookup.

Every shipped product (CardPointers, MaxRewards, point.me, Kudos) solves only the retrieval sub-problem over flat, hand-maintained rules tables. None of them plan, and none survive a mid-conversation change.

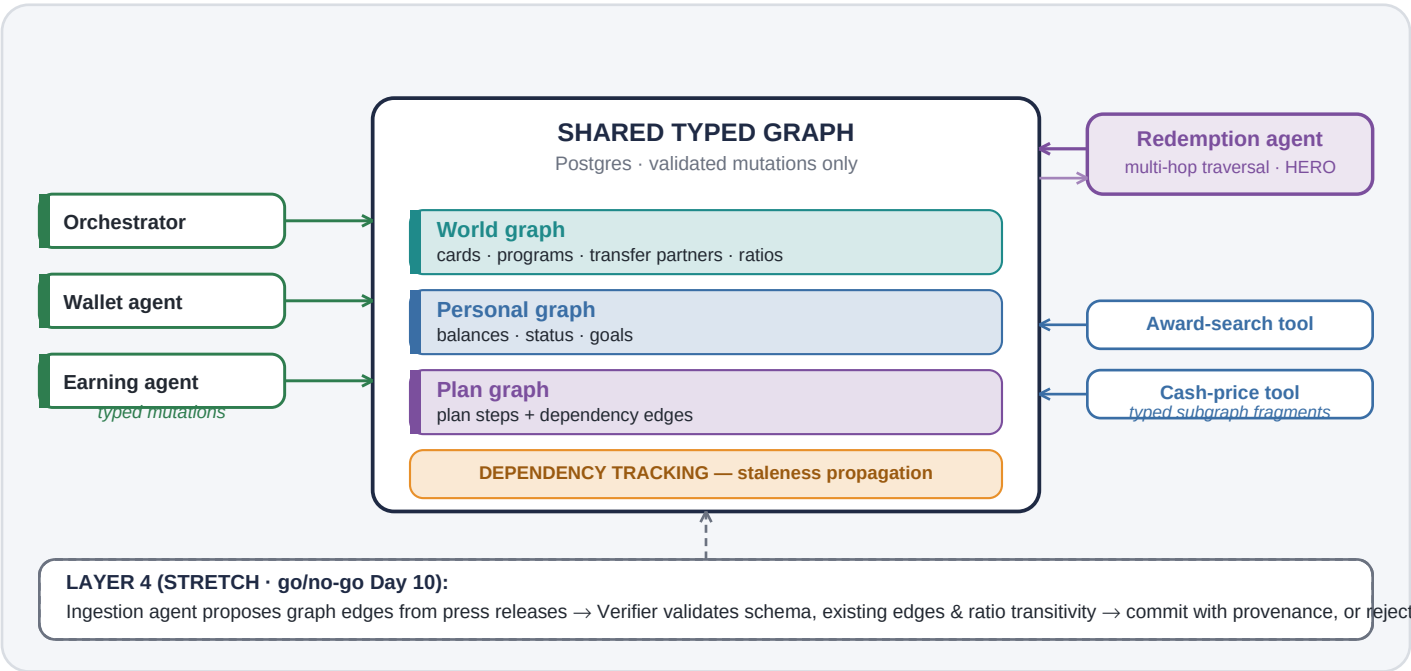
WHY THIS IS A BREAKTHROUGH

We treat agent coordination as shared typed state, not message-passing. That unlocks four behaviors no incumbent can produce:

- Portfolio-level multi-step plans with reasoning on every step.
- Plans that survive mid-conversation state changes via structural invalidation.
- Tool results that compose into the shared world model instead of being re-fetched.
- Domain facts learned from unstructured updates under verification, not by hand.

The lasting contribution is the benchmark: the first head-to-head evaluation of typed-graph vs. free-text coordination on a structured planning task.

ARCHITECTURE — COORDINATION AS SHARED STATE



A balance change → dependent plan steps marked stale → redemption agent re-plans automatically, with no orchestrator message.

SCOPE OF WORK

IN SCOPE — MVP

- Manually-entered wallet (demo persona: 5 cards, 240k pts, Tokyo in Oct).
- 20 pre-seeded cards + transfer-partner network (Chase UR, Amex MR).
- Layers 1-3 end to end; dependency tracking on plan nodes only.
- One real tool (cash-price); fixture-based award availability.
- 30-query benchmark + two well-tuned baselines.
- Layer 4 ingestion+verifier — hard-cuttable, go/no-go Day 10.

OUT OF SCOPE

- Plaid / live transaction ingestion.
- OAuth real-time issuer balance sync.
- Live award search across a full alliance.
- Mobile app.
- Full transitive dependency propagation.
- Production verifier with human review loop.

TEAM & HOW THE WORK DIVIDES

ALAN

Person A · Graph / Persistence

Schema, Postgres graph layer, dependency-tracking edges, optimistic concurrency, `serialize_world_graph()`. Critical path — unblocks every lane on Day 1.

VAL

Person B · Frontend / Demo

Graph-mutation sidebar, stale-node lighting, head-to-head contrast UI, demo polish. Makes the architecture visible to evaluators.

MICHAEL

Person C · Redemption / Eval

Redemption agent (hero), graph-typed tools, 30-query benchmark, BOTH baselines (single-agent + CrewAI free-text), Layer 4. Most loaded lane.

RAQ

Person D · Orchestrator (owner, lead)

Orchestrator, wallet + earning agents, integration surface, cross-lane unblocking, schema change control.

WHAT'S ON SCREEN DURING THE 10-MINUTE DEMO

0:00–2:00

Query + coordination

Persona set. NL query: best use of points for Tokyo. Agents coordinate; sidebar streams typed mutations; a multi-step plan appears with reasoning per step.

2:00–4:30

Hero 1 — live re-plan

"I transferred 60k Chase to Hyatt yesterday." Wallet mutates the balance; dependent plan steps light up stale; redemption agent re-plans automatically.

4:30–6:30

Hero 2 — learn under verification

(Layer 4, if landed) Paste a real transfer-bonus press release. Ingestion proposes an edge; verifier accepts with provenance or rejects a hallucinated one. Show ≥ 1 reject.

6:30–9:00

Head-to-head

Same scenarios vs. single-agent and free-text baselines. Baseline hallucinates a ratio, misses the invalidation, re-fetches a tool. Show the benchmark numbers.

9:00–10:00

Close

The contribution: the coordination primitive + a released benchmark. Wins by KIND on plan invalidation — baselines structurally cannot track it.

GATES

Day 7: end-to-end path + live re-plan working
Day 10: Layer 4 go/no-go · Jun 29: demo

HYPOTHESIS (pre-committed)

Typed-graph beats both baselines by >10pp on accuracy & hallucination; categorical win on plan-invalidity. Benchmark released regardless of outcome.